

## BE 301: Biomechanics

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|---------------|---------------------------------------------------------------|
| Course Code   | BE 301                                                        |
| Course Name   | Biomechanics                                                  |
| L-T-P-C       | 3-0-2-4                                                       |
| Prerequisites | Mechanics of Rigid Bodies (IC 240)                            |
| Intended for  | B. Tech M.Tech Integrated Dual Degree Bioengineering students |
| Approval      | recommended by the 38th BoA (Jan 2021)                        |

### Course Contents

- **Introduction and Fundamentals**

what is Biomechanics? Anatomical Concepts in Biomechanics, free-body diagrams and equilibrium; linear and angular kinematics, kinetic equation of motion, work and energy method, application to biological systems: stress, strain, Modulus, strain energy, tension, compression, torsion, bending, buckling. [14 Lectures]

- **Tissues**

Animal tissues and plant tissues. Classification of animal tissues - hard tissue, soft tissue, properties of plant and animal tissues according to mechanics view point, Structure, Function, composition, material properties and modeling of tissues, Plant tissues - vascular bundles - xylem and phloem. Continuum Mechanics Concepts in Modeling of large deformation, Finite Element Modeling. [14 Lectures]

- **Joints and Movements**

Classification of joints, forces and stresses, biomechanical analysis joints, Gait, Joint replacement and reasons, Finite Element Modeling. [8 Lectures]

- **Biofluid mechanics**

Flow properties of blood and others, Fluid flow in plants, Dynamics of fluid flow in the biological system - modeling and experimental approaches, Measurement/Estimation of In-vivo elasticity of fluid transporting vessels.[6 Lectures]

### Laboratory Modules

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- To determine the tensile properties of a material (root or bones or plants or others).
- To determine the bending properties of a material (root or bones or plants or others).
- To determine the hardness properties of a material.
- To determine the torsional/shear properties of a material (root or bones or plants or others).
- To determine the buckling properties of a material (root or bones or plants or others).
- To determine the energy absorbed and toughness of a material (root or bones or plants or others).
- To determine the wear properties of material and different combination of material
- To determine the ground reaction force during normal walking or running.
- Finite Element Modelling and analysis of hard tissue and soft tissue (examples: Bone, ligaments or muscles)
- Fluid flow through the cardiovascular system: Simple modelling and analysis

### Text Books

1. M. Nordin and V. H. Frankel, **Basic Biomechanics of the Musculoskeletal System**, Lippincott Williams and Wilkins, 2012.
2. Y. C. Fung, **Biomechanics: Mechanical Properties of Living Tissues**, 2nd edition, Springer, 2007.

### References

1. K. J. Niklas, **Plant Biomechanics: An engineering approach to plant form and function**, University of Chicago Press, 1992.
2. Ozkaya, Nordin, Goldsheyder and Leger, **Fundamentals of Biomechanics: Equilibrium, Motion, and Deformation**, 3rd Edition, Springer, 2014.
3. R. L. Huston, **Fundamentals of Biomechanics**, CRC Press, 2013.